

Mathematics for Machine Learning

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Definition

The series $\sum_{n=1}^{\infty} a_n$ is called convergent if A is finite, otherwise it is called divergent.

Theorem

If $\sum_{n=1}^{\infty} a_n$ is convergent, then $a_n \rightarrow 0$. The inverse is not true.

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Theorem

The series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if for every $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that for every natural $n \geq n_0$ and m holds

$$|a_{n+1} + \dots + a_{n+m}| < \varepsilon.$$

Theorem

If $a_n \geq 0$ for all $n \in \mathbb{N}$ then $\sum_{n=1}^{\infty} a_n$ is convergent or it is equal to $+\infty$.

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If $a_n, b_n \geq 0$ and $a_n \leq b_n$ for all $n \in \mathbb{N}$ then if $\sum_{n=1}^{\infty} b_n$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent too.

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Theorem

If $a_n, b_n \geq 0$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = K$, $0 \leq K < \infty$ then if $\sum_{n=1}^{\infty} b_n$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent too.

Theorem

Let $a_n \geq 0$ for all $n \in \mathbb{N}$ and $\overline{\lim}_{n \rightarrow \infty} \sqrt[n]{a_n} = K$. Then

- ① if $K < 1$ then $\sum_{n=1}^{\infty} a_n$ is convergent,
- ② if $K > 1$ then $\sum_{n=1}^{\infty} a_n$ is divergent.

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Theorem

Let $a_n \geq 0$ for all $n \in \mathbb{N}$. Then

- ① if $\overline{\lim}_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} < 1$ then $\sum_{n=1}^{\infty} a_n$ is convergent,
- ② if $\underline{\lim}_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} > 1$ then $\sum_{n=1}^{\infty} a_n$ is divergent.

Indefinite Integral

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Definition

The set of all antiderivatives of function f is called indefinite integral of f :

$$\int f(x) dx = F(x) + C.$$

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- ③ If the functions $f(t)$ and $\varphi'(x)$ are continuous on the intervals T and X respectively and $\varphi(X) \subset T$ then

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④ If $f, g \in C^1(X)$, then

$$\int f dg = fg - \int g df.$$

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Definition

Let $f : [a, b] \rightarrow \mathbb{R}$ and $\xi_i \in [x_i, x_{i+1}]$, $i = 0, \dots, n-1$. The sum $\sigma = \sum_{i=0}^{n-1} f(\xi_i) \Delta x_i$ is called Riemann integral sum.

Definition

We will say that σ tends to I , when $\lambda \rightarrow 0$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for any partition with diameter satisfying to $\lambda < \delta$ and for every Riemann integral sum σ corresponding to that partitions holds $|\sigma - I| < \varepsilon$.

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Theorem

If $f \in \mathcal{R}[a, b]$, then it is bounded. The inverse is not true.

Example

Dirichlet function is bounded but not integrable

$$D(x) = \begin{cases} 0, & x \in \mathbb{Q} \cap [0, 1], \\ 1, & x \notin \mathbb{Q} \cap [0, 1]. \end{cases}$$

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Taking $\xi_i \in \mathbb{Q}$ for all i , we will have $\sigma = 0$ and in the case of $\xi_i \notin \mathbb{Q}$ for all i we will have $\sigma = 1$.

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Remark

If X is not bounded from below (above), then we will denote $\inf X = -\infty$ ($\sup X = +\infty$).

Theorem

If $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function, then it is integrable if and only if

$$\lim_{\lambda \rightarrow 0} \sum_{i=0}^{n-1} \omega_i \Delta x_i = 0,$$

where $\omega_i = M_i - m_i$, $M_i = \sup_{x \in [x_i, x_{i+1}]} f(x)$, $m_i = \inf_{x \in [x_i, x_{i+1}]} f(x)$.

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If $f \in C(a, b)$ and f is bounded in $[a, b]$, then $f \in \mathcal{R}[a, b]$.